

Fighting Misinformation on Social Media: Why People Share Fake News and Whether Credibility Indicators Help

Shaojing Fan^{ID}, National University of Singapore

Bryan L. Koenig^{ID}, Southern Utah University

Nasir Memon^{ID}, New York University

Mohan S. Kankanhalli^{ID}, National University of Singapore

Social media has become a powerful force in shaping public opinion and political behavior worldwide.¹

The rapid spread of misinformation, further amplified by generative artificial intelligence (AI),² poses significant societal risks. Beyond automated detection algorithms,³ platforms increasingly deploy credibility indicators as governance tools embedded in their infrastructures.

Rather than functioning merely as interface nudges, these indicators signal institutional authority, allocate responsibility, and structure how users interpret information.⁴ Theoretically, credibility indicators can operate through dual cognitive pathways: as heuristic cues that trigger rapid, intuitive judgments, or as prompts that encourage more deliberate, analytic evaluation. Understanding which pathway dominates is essential for assessing their real-world effectiveness.

Diverse credibility indicators have been developed,⁵ and prior studies show their effects vary by political affiliation⁶ and preexisting attitudes.⁷ However, most research focuses on single regions and specific news domains, limiting generalizability. To address this gap,

Social media platforms deploy credibility indicators as governance tools to curb misinformation, yet their effectiveness remains unclear. Drawing on dual-process theory, we examine whether such indicators engage analytic processing or merely function as heuristic cues.

we conduct a study across diverse, digitally connected populations with distinct media environments to examine how credibility indicators function across varying contexts. We further account for user background factors, such as education level and social media behavior to capture differences in digital literacy and information practices.

We presented participants with 30 text-based news posts covering diverse topics, each accompanied by images or videos; 25 included a red credibility warning (see Figure 1). After each post, participants indicated whether they would share it, explained their reasoning, reported whether the warning influenced them, and completed a survey on social media use and online behavior. Unlike prior work, our design concurrently evaluates

news characteristics, credibility indicators, user behavior, and self-reported motivations.

Results show that credibility indicators had limited impact on sharing intentions overall, across warning sources and genres. Their effectiveness varied by user background, including education level and social media behavior. We also observed that greater concern about fake news and higher perceived utility of credibility indicators were associated with increased sharing intentions. Sharing decisions were driven more by perceived social relevance than veracity. Our findings suggest concrete implications for designing misinformation mitigation systems. Credibility indicators are most effective when integrated into platform architecture, for example, by prompting brief user reflection or source consultation at the moment of sharing. To prevent inadvertent amplification in high-engagement domains, such as health news, cues should be context-sensitive, distinguishing uncertainty, dispute, and verified falsehood. Finally, the differential impact of institutional versus AI or crowd-sourced labels highlights the importance of clearly signaling authority and transparency to maintain user trust.

RELATED WORK

Curbing misinformation on social media platforms

Misinformation has been a huge issue for governments and social media practitioners.⁸ Governments fight misinformation with regulations and policies.⁸ Researchers from academia and industry develop automated methods to curb misinformation.² Natural language processing (NLP) and deep learning methods are widely

used for automated detection of fake news.³ Some researchers use convolutional neural networks and attention mechanisms to extract relevant deep features.³ Others apply generative adversarial networks to generate event-invariant multimodal features or new data instances for misinformation detection.⁹ These automated approaches have achieved considerable accuracy. However, they are highly data dependent, and therefore vulnerable to novel contexts. Due to the complex and subjective nature of misinformation, without a comprehensive understanding of human cognition, these algorithms struggle to generalize to new data and social situations.⁸

User nudging for misinformation

User nudging is another countermeasure to misinformation.¹⁰ Diverse applications and analyses have been proposed to remind and advise users against fake news, and one important type being credibility indicators.⁵

Previous literature shows a significant variation in the effectiveness of misinformation nudging. Some report clear effectiveness.⁶ Koch and colleagues¹¹ found that warning labels reduced both credibility perceptions of a fake news post on climate change and amplification intentions (that is, liking and sharing), whereas removing social endorsement cues below a post had no impact. The effectiveness of misinformation nudging seems to be influenced by multiple factors. Nekmat¹² reported the impact from news sources—news sharing likelihood was overall lower for nonmainstream news than mainstream news, but results showed a greater decrease for mainstream news when nudged. Hameleer and Meer¹³ found that for political news, fact checkers are most impactful for users when they confirm prior attitudes and less so when incongruent, indicating a problematic confirmation bias. In our work, we extend this research on text news by having news



Despite a minimum enlistment age of 18, large numbers of boys have been recruited into the Myanmar's national army for the country's internal conflicts.



Warning: Computer algorithms using Artificial Intelligence (AI) techniques dispute the credibility of this news.

Suppose you see the above news online, please answer the following questions.

Q1. Would you share this piece of news with your friends on social media?

☐ Yes
☐ No

FIGURE 1. User interface of our human study.

with both textual and visual information. We also use a broader selection of news topics and more variants of nudges (that is, credibility warnings with different attributed sources).

METHODOLOGY

Approval and consent

The experiment was approved by the Institutional Review Board of the National University of Singapore. A participant information sheet was presented on the first page of the experiment, and informed consent was obtained prior to participation.

Participant recruitment

Our final sample was 272 workers from the online crowdsourcing platform Amazon Mechanical Turk (MTurk), 105 participants from India [73.3% male, age: mean = 32.70, standard deviation (SD) = 4.96] submitting 105 surveys, and 167 participants from the United States (51.5% male, age: mean = 36.20, SD = 9.28) submitting 167 surveys. All participants reported normal or corrected-to-normal color vision. We chose two countries of residence (India and United States) as they are two

large English speaking countries with different cultural backgrounds and government policies. They are also the two most common residential places for MTurk workers. Studying these two groups of people provides insights on how user background influences the sharing and curbing of misinformation. Our 272 participants reflect a diverse demographic profile across ethnicity, age, and education, with the sample primarily comprising White (47%) and Asian/Pacific Islander (40%) participants, and a majority aged 25–34 (57%) holding a bachelor's degree (68%). Detailed demographic breakdowns are provided in the supplementary materials.¹

Stimuli collection

We compiled a stimulus data set comprising 76 unique news posts, each consisting of a headline paired with either a static image or a short video clip [see Figure 2(a)–(c) for examples]. Of these, 30 were true news posts and 46 were fake news posts. True news headlines were selected from established news outlets, for example, official news websites, such as BBC, CNN, Reuters, The Guardian, and Times of India. The

authors verified the accuracy of these headlines through cross-platform validation, prioritizing posts reported by multiple independent sources.

Fake news posts were created in three ways: 1) *headline fabrication*—inaccurate headlines paired with real visual content; 2) *misrepresentation*—true headlines paired with irrelevant but real visual content, that is, images or videos taken out of context; and 3) *complete fabrication*—both untrue headlines and fake visual content generated or altered using computational methods, such as deepfakes¹⁴ [see Figure 2(a)–(c)]. Videos for fake posts were either manipulated from legitimate sources or selected from the YouTube-8M data set¹⁵ to ensure ecological validity and realistic exposure. Each video clip ranged from 3 to 6 seconds and depicted scenes relevant to the headline.

The news posts covered diverse topics. The number of posts per topic was as follows: politics (11 fake, eight true), health and life science (nine fake, four true), entertainment and leisure (20 fake, seven true), and disasters (six fake, 11 true). For each news post, we also created a matching version with a static image extracted from the video.

Beijing protester shot in chest; six live rounds fired during China National Day rally.



(a)

Two people killed during a California school shooting on Thursday (Nov 14), a teen girl in custody.



(b)

Obama calls President Donald Trump "a total and complete dipshit".



(c)

FIGURE 2. Examples of three types of fake news items based on how they were made. (a) Headline fabrication—fake news headline, true visual content; (b) misrepresentation—true news headline but an irrelevant visual content; (c) complete fabrication—both news headline and visual content are fake, with visual content generated by deepfake.¹⁴ For each news headline with a supporting image, we have a corresponding news headline with a short video clip showing the same content.

The extracted frame was selected as the one with the largest image entropy, computed as

$$S = \sum_{i=1}^n (-P_i \log_2 P_i),$$

where P represents a histogram with $n = 256$ bins for a grayscale image. The frame with the largest entropy serves as the most representative frame of the video clip. Given that our video clips depicted mostly consistent content across frames, these isolated frames closely represent the source videos.

The data set systematically varied news veracity, genre, fake news type, and media format (image versus video). Each item included only a headline and visual (no post text), ensuring participants focused on the combined headline-visual stimuli. We removed sound from all video stimuli to maintain experimental control and avoid confounds between video and image conditions as well as variability due to device settings and cross-national differences. However, this choice excludes important credibility cues (for example, tone, background sounds, narration style), meaning our stimuli represent a simplified approximation of real-world news consumption. Findings should therefore be interpreted with this limitation in mind.

Design and procedure

Our experiment design was based on that of Yaqub and colleagues.⁶ We collected data on MTurk, an online crowdsourcing platform. Each MTurk worker viewed and responded to 30 news posts sequentially in random order. The news posts were either headline with an image or a video. For news posts with a video, the video clip was played automatically once loaded onto the screen. Participants could replay the video as

many times as they wished. For each participant, a red credibility warning was displayed below the 25 news posts, and the other five did not have a credibility warning. The five items presented without a credibility indicator per participant were randomly selected and counterbalanced across participants from the full stimulus pool, ensuring that no fixed subset of posts was systematically assigned to the undisputed condition. This randomization of post order, indicator assignment, and media format for each participant reduced systematic confounding, helping ensure that observed effects reflect the manipulated variables rather than incidental stimulus properties.

To mitigate potential fatigue or desensitization effects from repeated exposure, post order and warning assignment were randomized at the participant level, preventing warnings from appearing in a predictable sequence. In addition, exposure length was self-paced, allowing participants to control their viewing time. This design reduces the likelihood that any observed effects are attributable to systematic ordering or cumulative exposure to credibility warnings rather than to the manipulated content itself.

Each warning message was randomly assigned one attributed source from these four:

- › *Fact checkers*: Multiple fact-checking journalists dispute the credibility of this news.
- › *News media*: Major news outlets dispute the credibility of this news.
- › *Public*: A majority of social media users disputes the credibility of this news.
- › *AI*: Computer algorithms using AI techniques dispute the credibility of this news.

In defining “AI algorithm” as a source, we acknowledge the ongoing discourse regarding the reliability and perceived validity of such automated systems. Following recent calls for the standardization of AI and sensor technology assessments to ensure consistent trust and accuracy,¹⁶ our study utilizes AI as a distinct source to evaluate how users’ trust is modulated by the perceived (un)standardized nature of algorithmic authority.

After viewing each news post, participants indicated whether or not they would share the corresponding news on social media. They then indicated why. For each news post, participants indicated how influential they thought the credibility warning was on their decision, choosing from “no influence,” “some influence,” and “great influence.” At the end of the study, questions asked participants about demographics and their perceptions of social media and habits on it. All measures can be found in the supplementary materials available at <https://doi.org/10.1109/MC.2026.3686119>, provided by the authors.

To ensure data quality, we implemented several screening criteria: 1) participants were required to have a Human Intelligence Task (HIT) approval rate of $\geq 95\%$ with at least 100 previously approved HITs; 2) we excluded participants who completed the survey unusually quickly, within 300 seconds, to ensure thoughtful responses; 3) we excluded any repeat participants by tracking MTurk Worker IDs, retaining only their first survey completion. These measures helped us maintain data integrity by minimizing inattentive or duplicate responses, ensuring that the data set reflects reliable and independent participant input.

The final sample of 272 respondents provided 7,570 responses (3,199 true news; 3,424 share intents; 5,200 disputed news posts^a). We refer to the final sample throughout the article.

Analytical approach

We selected statistical methods appropriate for each outcome type. For binary outcomes (for example, share versus not share), we used generalized linear mixed models (GLMMs), a regression approach suitable for nested data. GLMMs were used to account for the nonindependent structure of multiple posts per participant and to model binary outcomes. Including participant-level random intercepts allows the model to separate individual differences in sharing tendency from the effects of experimental manipulations. Results are reported as odds ratios (OR), indicating how much more (or less) likely sharing becomes when a predictor is present; for example, OR = 1.50 means 50% higher odds of sharing. We also report 95% confidence intervals (CIs) to convey estimate precision. Interaction effects were tested using likelihood-ratio tests, and pairwise comparisons applied Bonferroni correction to control for multiple testing.

For continuous or proportion outcomes aggregated at the post level (for example, mean sharing intent), nonparametric Wilcoxon signed-rank tests were applied due to bounded distributions (0–1), reporting W , p , and effect size $r = |Z|/\sqrt{N}$. User traits were analyzed with generalized estimating equations (GEEs) to estimate population-averaged effects while

accounting for repeated measures, and exploratory linear regressions predicted sharing of disputed news controlling for baseline behavior.

A power analysis evaluated the ability to detect a Country $\times$ HasIndicator interaction. At the participant level (United States: 167, India: 105), power was low for small to medium effects (0.13–0.36) but adequate for large effects (0.98). Adjusting for clustering (effective $N \approx 944$ United States, 631 India; intraclass correlation coefficient (ICC) = 0.1, design effect = 2.16) increased power for medium to large effects (0.97–1.00) while remaining low for small effects (0.49). Nonsignificant interactions should thus be interpreted cautiously.

The primary outcome was sharing intent (1 = share, 0 = not share), and the main predictor was the presence of a credibility indicator (1 = disputed, 0 = undisputed). Fake news type was classified as headline fabrication, misrepresentation, or deepfake, with true news as the reference. GLMMs with participant-level random intercepts were used to account for the nested data structure and ensure fixed-effect estimates reflect within-person responses rather than between-person differences. Randomization of post order, indicator assignment, and media format helped minimize systematic confounding across stimuli and conditions.

RESULTS

Limited effectiveness of credibility indicators in reducing fake news sharing

Credibility indicators did not reduce sharing overall, and their effects were inconsistent across indicator source types. The binary presence of

a credibility warning indicator had no significant effect on participants' sharing decisions [OR = 0.86, 95% CI (0.52, 1.41), $z = -0.61$, $p = .544$], suggesting that warning labels alone are insufficient to deter sharing.

No significant differences emerged among the four indicator source types in pairwise comparisons (all $p > .28$). Only fact checker (OR = 0.68, 95% CI [0.51, 0.89], $p = .006$) and news media [OR = 0.69, 95% CI (0.52, 0.92), $p = .012$] differed significantly from the no-indicator baseline, whereas AI [OR = 0.83, 95% CI (0.62, 1.12), $p = .221$] and public [OR = 0.82, 95% CI (0.62, 1.10), $p = .190$] did not.

To examine whether the effect of credibility indicators on sharing intent varied across news topics, we conducted Wilcoxon signed-rank tests comparing mean sharing intent between undisputed (no indicator) and disputed (indicator present) conditions for each news genre (politics, health, entertainment, and disaster), separately by news veracity (true versus fake) and in aggregate. Table 1 presents mean sharing intent (%) under each condition alongside associated p -values. Across all 100 posts, the presence of a credibility indicator significantly increased sharing intent overall (undisputed: 42.8% versus disputed: 49.6%; $W = 1405$, $p < .001$, $r = .39$), driven primarily by fake news posts (40.9% versus 50.2%; $W = 357$, $p < .001$, $r = .49$), whereas the effect for true news posts was nonsignificant (45.2% versus 48.8%; $p = .158$). In further analyses at the genre level, significant indicator effects emerged for health news in aggregate (31.6% versus 46.2%; $W = 13$, $p = .005$, $r = .69$) and for disaster news (43.2% versus 49.3%; $W = 148$, $p = .029$, $r = .38$). Within health news, the effect was significant for fake health

^aIn this article, "disputed news posts" and "undisputed news posts" refer to news posts that come with or without credibility indicator, respectively.

posts specifically (24.1% versus 40.6%; $W=5, p=.039, r=.69$), as was the disaster-fake news comparison (40.2% versus 50.5%; $W=13, p=.021, r=.63$).

Notably, these effects were consistently positive. That is, the presence of a credibility indicator increased rather than decreased sharing intent, suggesting a potential backfire effect whereby disputed labels may heighten perceived salience or newsworthiness of the flagged content, particularly for health and disaster topics. In contrast, politics and entertainment genres showed no significant indicator effects across either veracity condition (all $p > .08$), indicating that the influence of credibility indicators on sharing intent is topic-dependent.

These findings suggest that the effectiveness of credibility indicators depends on the perceived authority of their source, with professional and institutional labels exerting a modest suppressive effect on sharing that

algorithmically or crowd-sourced labels do not. The effect is also related to the news topics, with indicators increasing sharing intent for health and disaster content, and especially for fake news, while showing no significant influence in politics and entertainment, suggesting a topic-dependent and, in some cases, backfire effect.

Higher concern about fake news and greater perceived utility of indicators are associated with increased sharing

To investigate how individual differences relate to sharing behavior, we examined two user-reported factors: concern about fake news and perceived utility of credibility indicators, using GEEs to account for repeated measures across posts.

Greater concern about fake news predicts higher sharing: The analysis revealed a significant positive association between concern about fake

news and the likelihood of sharing posts. Specifically, each one-unit increase in concern corresponded to a 0.477 increase in the log-odds of sharing ($z = 7.23, p < .001$). This effect remained significant when controlling for the presence of a credibility indicator and advertisement type (coefficient = 0.468, $p < .001$). In the controlled model, the credibility indicator did not significantly predict sharing ($p = 0.631$), while advertisement type showed only a marginal effect ($p = 0.071$). These results indicate that participants with higher concern about fake news were consistently more likely to share posts, independent of indicator presence or ad type.

Higher perceived utility of credibility indicators predicts increased sharing: Similarly, participants who rated credibility indicators as more useful were more likely to share news content. Each one-unit increase in

TABLE 1. Mean sharing intent (%) for news posts by news genre and veracity, comparing undisputed (no credibility indicator) versus disputed (indicator present) conditions. Ninety-five percent CIs were computed via bootstrap resampling (5,000 iterations). p -values are from Wilcoxon signed-rank tests (paired by post). Statistically significant effects ($p < .05$) are shown in boldface.

	True news					Fake news					Total (true and fake)				
	Undisputed		Disputed			Undisputed		Disputed			Undisputed		Disputed		
Genre	%	95% CI	%	95% CI	<i>p</i>	%	95% CI	%	95% CI	<i>p</i>	%	95% CI	%	95% CI	<i>p</i>
Politics	43.1	[31.9, 54.2]	44.6	[40.3, 49.4]	.844	43.5	[34.6, 51.8]	50.8	[45.2, 55.9]	.083	43.3	[36.3, 49.8]	48.2	[44.2, 52.2]	.210
Health	43.0	[33.0, 53.0]	54.6	[44.6, 62.0]	.062	24.1	[11.0, 39.5]	40.6	[36.1, 45.6]	.039	31.6	[21.5, 42.3]	46.2	[40.5, 52.2]	.005
Ent.	48.1	[39.1, 56.8]	49.3	[41.7, 56.4]	.922	46.5	[38.3, 54.3]	53.3	[48.9, 57.6]	.097	47.0	[40.5, 53.0]	52.1	[48.3, 55.8]	.118
Disaster	45.3	[39.0, 51.7]	48.5	[43.0, 54.1]	.465	40.2	[30.9, 49.3]	50.5	[45.2, 56.0]	.021	43.2	[37.8, 48.5]	49.3	[45.3, 53.4]	.029
Total	45.2	[40.8, 49.6]	48.8	[45.3, 52.4]	.158	40.9	[35.7, 46.1]	50.2	[47.4, 53.0]	<.001	42.8	[39.3, 46.4]	49.6	[47.5, 51.7]	<.001

Ent: entertainment; undisputed: posts without a credibility indicator; disputed: posts with a credibility indicator present. Ninety-five percent CIs reflect bootstrap uncertainty around the post-level mean sharing intent ($N = 100$ posts total; true news $n = 43$, fake news $n = 57$). CI width reflects the number of posts per genre-veracity cell, which varies from 6 to 34. Postlevel paired analyses.

perceived utility corresponded to a 0.209 increase in the log-odds of sharing ($z = 3.39$, $p = 0.001$). This effect also remained significant after

controlling for the credibility indicator and advertisement type (coefficient = 0.212, $p = 0.002$), whereas neither the indicator ($p = 0.360$) nor

ad type ($p = 0.978$) significantly influenced sharing.

Overall, these findings demonstrate that higher user concern about fake news and greater perceived usefulness of credibility indicators are positively associated with sharing behavior, regardless of indicator presence or advertisement type.

TABLE 2. Regression β coefficients for individual difference variables predicting sharing likelihood of disputed news posts, first alone (zero-order correlations) and then controlling for participants' sharing likelihood of undisputed news posts. Individual differences variables are excluded if they did not correlate significantly with rates of sharing disputed news. * $p < .05$, ** $p < .01$, *** $p < .001$.

Explanatory variables	β (correlation)	β controlling for % undisputed
Concerns about security and fake news		
Concern about data breaches	.34***	.05
Fake news concern impacts social media	.33***	.08
Fake news concern impacts news sharing	.19***	.05
Online behaviors		
Frequency share news on social media	.51***	.19***
Years on social media	-.48***	-.19***
Recommend social media	.43***	.07
Number of social media friends	-.24***	-.11**
Frequency of using chat	.27***	.08*
Daily social media use time	.46***	.14**
Social media app usage		
Twitter	.19***	.03
Instagram	.13***	.08*
Snapchat	.36***	.17***
WhatsApp	.64***	.25***
LinkedIn	.39***	.19***
WeChat	.53***	.37***
Weibo	.55***	.36***
Tumblr	.36***	.25***
Pinterest	.25***	.18***

Influence of user background

We conducted exploratory analyses on individual difference measures to see which would predict sharing disputed news posts. To do so, we calculated for each participant the sharing likelihood of undisputed news posts and disputed news posts. This allowed us to control for intent to share undisputed news in order to see what predicts the intent to share disputed news above and beyond the general tendency to share news.

Surprisingly, neither concern about fake news nor concern about data breaches predicted the intent to share fake news, but social media use and online behaviors did. As shown in Table 2, variables related to data security and fake news did not robustly impact intent to share disputed news. By contrast, online behaviors and social media app use did predict the intent to share disputed news—even when controlling for the intent to share undisputed news. The strongest predictors of the intent to share disputed news are the usage of specific apps, such as Whatsapp, followed by user online behaviors, such as frequency of sharing news on social media, and daily social media use time.

GLMM analyses revealed that education was positively associated with sharing behavior: each one-unit increase in education corresponded to an 84% increase in the odds of sharing

[$\beta = 0.61$, $SE = 0.14$, $z = 4.36$, $p < .001$; $OR = 1.84$, 95% CI (1.40, 2.42)]. Importantly, this effect was consistent across both Indian and U.S. participants, as indicated by a nonsignificant Education $\times$ Country interaction ($\beta = 0.16$, $SE = 0.20$, $p = .424$). The presence of a credibility indicator did not significantly influence sharing [$\beta = -0.16$, $SE = 0.26$, $p = .535$; 0.85, 95% CI (0.51, 1.42)], suggesting that educational background may shape sharing decisions independently of platform-based credibility cues.

Social motivation overwhelmed truth concerns

To examine why participants chose to share or not share news posts while accounting for repeated observations within participants, we analyzed reason-selection indicators using GLMMs with a binomial outcome and participant-level random intercepts. Figure 3(a) and (b) summarizes the overall frequency of each reason. Each binary reason indicator was modeled with the presence of a credibility indicator (HasIndicator) as the key predictor, with separate models for nonsharing ($N = 4,001$) and sharing decisions ($N = 3,369$).

Among the 4,001 nonsharing decisions, the most frequently cited reasons were “not relevant to my life” (29.5%), “influenced by the red warning” (14.9%), and “no discussion among friends” (13.7%). The presence of a credibility indicator significantly predicted eight of nine nonsharing reasons. Indicators increased the odds of citing distrust of the headline [$OR = 2.44$, 95% CI (1.69, 3.53), $p < .001$], distrust of the image or video [$OR = 2.21$, 95% CI (1.53, 3.19), $p < .001$], and the red warning itself as a deterrent [$OR = 65.01$, 95% CI (16.05, 263.23), $p < .001$],

while simultaneously reducing the odds of citing irrelevance [$OR = 0.46$, 95% CI (0.37, 0.57), $p < .001$], disinterest [$OR = 0.58$, 95% CI (0.44, 0.77), $p < .001$], and lack of discussion value [$OR = 0.64$, 95% CI (0.50, 0.82), $p < .001$]. This pattern suggests that while the indicator successfully activates credibility-related concerns, these concerns primarily operate by displacing social and interest-based motivations rather than eliminating them.

Among the 3,369 sharing decisions, the indicator significantly reduced the odds of citing agreement [$OR = 0.58$, 95% CI (0.44, 0.75), $p < .001$] and personal relevance [$OR = 0.43$, 95% CI (0.33, 0.56), $p < .001$], indicating that even when credibility concerns are present, participants may still proceed to share when underlying social motivations, such as perceived relevance or engagement value, remain sufficiently strong.

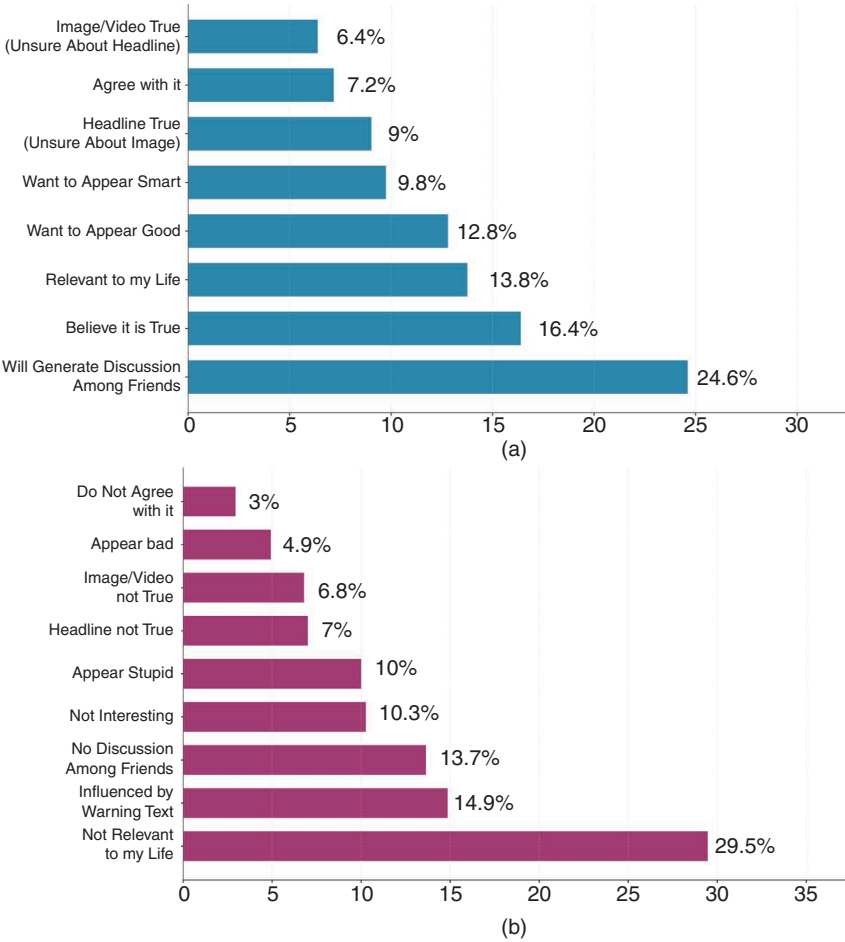


FIGURE 3. (a) Percentage of intend-to-share news posts for which participants chose each reason for sharing (n = 3,369). (b) Percentage of not-intend-to-share news posts for which participants chose each reason for not sharing (n = 4,001).

Effects of fake news type and credibility indicators on sharing

To examine how fake news type affects sharing behavior and whether credibility indicators moderate this effect, we fitted a series of GLMMs with a binomial outcome (share = 1) and a participant-level random intercept, using true news (FakeType = 99) as the reference category and no indicator (undisputed) as the reference for the credibility indicator factor.

The overall random-intercept variance was substantial ($\sigma^2 = 5.05$, ICC = 0.61), confirming that a large proportion of outcome variance was attributable to stable individual differences in sharing propensity. Among the fake news types, deepfakes were shared significantly less often than true news [OR = 0.56, 95% CI (0.46, 0.68), $z = -5.76$, $p < .001$], indicating that participants were more skeptical of manipulated audio-visual content, whereas headline fabrication news (authentic visual paired with fabricated text) did not differ significantly from true news in sharing likelihood (OR = 0.89, $p = .125$). Interestingly, misrepresentation news (authentic text paired with a fabricated or misrepresentative image/video) elicited higher sharing rates than true news [OR = 1.23, 95% CI (1.01, 1.50), $z = 2.01$, $p = .044$], suggesting that visually compelling but misleading imagery may enhance rather than inhibit sharing. To assess whether perceived visual quality contributed to sharing differences, we analyzed self-reported reasons for not sharing. Among fake news nonsharing decisions ($n = 2,394$), visual distrust was highest for deepfakes (16.4%; 105/642), followed by misrepresentation (12.7%; 58/455) and headline fabrication news (10.0%; 130/1,297). These differences were significant ($\chi^2(2) = 16.17$, $p < .001$).

Pairwise tests showed higher distrust for deepfakes than headline fabrication news [$\chi^2(1) = 15.58$, $p < .001$], but not compared to misrepresentation [$\chi^2(1) = 2.46$, $p < .117$] or between misrepresentation and headline fabrication [$\chi^2(1) = 2.33$, $p < .127$]. This suggests deepfakes may have been perceived as less realistic, potentially contributing to lower sharing rates.

Pairwise contrasts confirmed that all three fake news types were mutually distinguishable: misrepresentation news was shared significantly more than headline fabrication news (OR = 1.38, $p = .003$), and deepfakes were shared far less than both misrepresentation (OR = 0.46, $p < .001$) and headline fabrication news (OR = 0.63, $p < .001$). Within this model that adjusts for fake news type, the coefficient for the credibility indicator was positive [OR = 1.66, 95% CI [1.41, 1.96]], consistent with the backfire pattern observed in the genre-level analyses. As noted earlier, the indicator's overall marginal effect was not statistically significant (OR = 0.86, $p = .544$); the positive estimate here likely reflects the within-type association after controlling for baseline differences across content categories. This effect was most pronounced for deepfakes, where the undisputed-to-disputed shift in sharing rate was largest in absolute magnitude [+17.9 percentage points (pp), from 24.8% to 42.7%; $\chi^2(1) = 17.05$, $p < .001$], compared to +7.8 pp for headline fabrication news and +12.9 pp for misrepresentation news, though the FakeType $\times$ HasIndicator interactions in the full model were jointly nonsignificant [likelihood-ratio test (LRT): $\chi^2(3) = 3.64$, $p = .303$], indicating that the backfire effect did not significantly differ in magnitude across fake news types.

DISCUSSION

The present findings suggest that the limited effectiveness of credibility indicators is not simply a failure of design, but a reflection of a deeper misalignment between epistemic interventions and the social logic of platformed communication. Rather than treating sharing as a problem of truth evaluation alone, our results support framing news sharing as a form of social coordination and visibility management. Across analyses, credibility indicators did not reliably suppress sharing, and in several contexts, they were associated with increased sharing, a pattern more consistent with salience amplification than corrective nudging. This challenges a straightforward application of Nudge theory, which assumes that small informational cues can realign behavior with users' preexisting accuracy goals. Our data instead indicate that accuracy goals may be secondary to social incentives, such as discussion generation, identity signaling, and relevance maintenance. This interpretation aligns with prior work demonstrating that social motivations frequently outweigh epistemic concerns in digital environments (for example, CNN).¹⁷

Findings that initially appear paradoxical become theoretically coherent under this framing. Participants who reported greater concern about fake news were more likely to share disputed content. Rather than indicating low self-awareness, this pattern is consistent with several possible explanations, including attitude-behavior gaps, motivated reasoning, or heuristic processing under cognitive load. Participants who report high concern about misinformation may nonetheless rely on fast, intuitive judgments

when making sharing decisions, particularly under conditions of fatigue or attentional depletion that are common in extended digital tasks. We cannot definitively distinguish between these mechanisms based on self-reported data alone. Concern about misinformation may function as performative norm signaling, overconfidence in personal discernment (“I can tell what’s fake”), or endorsement of the belief that sharing does not necessarily imply agreement. Alternatively, the disconnect between expressed concern and sharing behavior may reflect cognitive resource depletion: participants completing a 30-item evaluation task may have increasingly relied on heuristic shortcuts rather than deliberative processing, regardless of their stated attitudes toward misinformation. In high-visibility social media environments, expressing concern and engaging with contested content are not mutually exclusive behaviors. Thus, concern does not automatically translate into restraint when social rewards (for example, attention, engagement, or group belonging), are salient.

Our findings also suggest that credibility indicators function not merely as neutral interface elements but as platform-embedded signals of authority. Warnings attributed to institutional actors (fact checkers and news media) modestly reduced sharing, whereas AI-generated and crowd-sourced warnings did not. This pattern reflects trust asymmetries across authority sources and supports interpreting indicators as governance tools rather than simple user interface nudges. From this perspective, the effectiveness of a credibility cue depends on how audiences attribute responsibility within the misinformation ecosystem.

Prior research shows that users differ in whether they believe platforms, journalists, governments, or citizens should act to curb misinformation.¹⁸ Our results suggest that institutional authority retains some persuasive power, but that algorithmic authority may not yet command equivalent legitimacy.

Importantly, the broader pattern indicates that credibility cues operate within an engagement-driven incentive structure. In health and disaster genres, indicators increased sharing, reflecting a backfire or salience effect. Drawing on risk mitigation frameworks from health and sports medicine, such as anticipatory planning lessons from the COVID-19 pandemic, these findings suggest that warnings may heighten perceived importance or controversy, reinforcing that sharing decisions are shaped more by platform visibility incentives than by adherence to factual accuracy.¹⁹

Among the three fake news types, participants were most likely to share misrepresentation (true headlines paired with misleading visuals), while deepfakes were shared least often. This pattern may suggest that users prioritize headlines over visuals when judging credibility. However, an alternative explanation is that deepfakes were perceptually less realistic. Supporting this, visual distrust was reported more frequently for deepfakes than for the other fake news types, with significant overall differences. Pairwise comparisons indicated higher distrust for deepfakes than headline fabrication but not compared to misrepresentation. Without independent realism ratings, we cannot disentangle these explanations and therefore present both interpretations. We recommend that future

studies include independent perceptual realism ratings to clarify the underlying mechanisms. Regardless of the mechanism, partially authentic misinformation appears especially difficult to counter. Misrepresentation was shared at rates comparable to true news, suggesting that users may overlook visual inconsistencies when headlines appear credible. In contrast, lower sharing of deepfakes may reflect greater skepticism, detectable artifacts, or both. These findings highlight the importance of controlling for stimulus quality in future work and suggest that misrepresentation may remain particularly challenging to detect even as deepfake realism improves.

Our research has several limitations. Prior research suggests the importance of political affiliation, especially in the United States.¹³ Political affiliation is clearly important for region-specific studies. Our study did not measure political affiliation because our participants are not limited to U.S. residents. Relatedly, our stimuli are not limited to U.S. news and political topics. Second, the news source itself can influence the effectiveness of news credibility indicators.¹² This is an important factor, but we decided to control for it to simplify our experiment’s design. Our news posts did not have the full post text, only a headline and visual. Moreover, our participants’ choices might have been influenced by noting that credibility warnings were showing up with true news, potentially undermining their trust of these indicators.

While our study offers controlled insights into credibility indicators, several methodological tradeoffs limit its generalizability. Specifically, our use of MTurk workers may have skewed the

sample toward a tech-savvy, English-literate demographic rather than reflecting a fully representative population, particularly among participants from India. While we implemented multiple screening measures to enhance data quality, it is possible that participants' responses were influenced by demand characteristics, such as perceived expectations about sharing behavior. Future work can consider alternative experimental designs or incentive structures to further mitigate this potential bias and confirm the robustness of observed sharing patterns. Additionally, our design presented credibility indicators on 25 of 30 posts (83%), a rate higher than typical social media environments where flagged content is relatively rare. This imbalance may have induced banner blindness or desensitization, leading participants to discount frequent warnings. Consequently, the observed limited efficacy of credibility indicators should be interpreted with caution and may not generalize to more naturalistic settings where warnings are less frequent and potentially more salient.

Furthermore, the use of brief (3–6 s) video clips without audio, while necessary for experimental control, reduces ecological validity. In real-world social media contexts, auditory cues are important for credibility assessment and engagement. Their removal limits our stimuli to a form of “thin-slice” visual processing that may not fully reflect naturalistic news consumption. Future work should examine how audio interacts with visual and textual cues, and whether these findings generalize to richer, more realistic settings. Finally, although we measured perceived influence of indicators and sharing reasons via self-report, these

may reflect post-hoc rationalization rather than true causal awareness. Participants may construct explanations after decisions, overstating credibility concerns while underreporting less socially desirable motivations (for example, entertainment value or emotional reactivity). Future research should use behavioral or experimental approaches, such as process-tracing and response time analyses, to better distinguish perceived from actual influences. Additionally, deeper psychological factors, such as smartphone dependency may also shape digital decision making and warrant further study.

Additionally, we did not measure transient cognitive or physiological states that may influence decision making. Factors, such as fatigue, stress, and cognitive load, can increase reliance on heuristic processing and reduce deliberation. Given that participants evaluated 30 news posts sequentially, fatigue may have accumulated, potentially shifting decisions toward more automatic responses. Future work should incorporate measures, such as cognitive load, time-on-task, or fatigue indicators (for example, self-reports, time of day) to better distinguish motivated reasoning from resource-related effects.

Nevertheless, our findings have concrete implications for designing misinformation mitigation systems and broader platform governance. Effective strategies should integrate credibility indicators into the sharing architecture rather than post-hoc labels, using friction mechanisms (for example, prompts to reflect, summarize, or consult sources) to guide behavior while preserving user agency. Platforms should employ tiered, context-sensitive cues to distinguish uncertainty,

dispute, and verified falsehood, and clearly communicate authority and responsibility: institutional actors signal evaluative roles, while algorithmic systems remain transparent and explainable.²⁰ Current interventions include algorithmic downranking, friction prompts, fact-checking partnerships, and community moderation, each with tradeoffs.²⁰ Because social incentives strongly drive sharing, mitigation should incorporate accountability mechanisms, such as visible correction histories, delayed virality thresholds, or reputational prompts, to align engagement with epistemic and civic responsibility. Overall, layered approaches combining visibility reduction, behavioral prompts, authoritative signaling, and user empowerment may be most effective, and future research should examine whether these strategies interact additively, synergistically, or in conflict.

Overall, our study found that credibility indicators had limited impact on sharing behavior. We selected India and the United States as large, English-proficient yet culturally distinct democracies with contrasting media ecosystems to examine how institutional trust and digital literacy moderate indicator effectiveness. Individual differences, such as education and social media app usage strongly predicted sharing intent, underscoring the need to move beyond one-size-fits-all interventions. The counterintuitive positive association between fake news concern and sharing suggests that epistemic cues may be overridden by social motivations or perceived duty to warn others. These findings suggest that credibility labeling could be enhanced by personalizing indicators

to users' social circles and communication habits. Future governance strategies should treat indicators not as isolated correctness signals but as part of a broader accountability ecosystem integrated with platform architectures and digital literacy initiatives.

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ABOUT THE AUTHORS

SHAOLING FAN is a senior lecturer in the Department of Electrical and Computer Engineering, National University of Singapore, Singapore 119077, Singapore. Her research interests include computer vision, cognitive vision, and computational social science. Fan earned her Ph.D. in communication and information systems from Ningbo University. Contact her at dcfsf@nus.edu.sg.

BRYAN L. KOENIG is an associate professor in the psychology department at Southern Utah University, Cedar City, UT 84720 USA. His research interests include social and evolutionary psychology, especially as related to morality and social relationships, with a developing interest in suicidality. Koenig earned his Ph.D. in social psychology from New Mexico State University. Contact him at bryankoenig@suu.edu.

NASIR MEMON is a professor at the New York University Tandon School of Engineering computer science and engineering department and affiliate faculty at the computer science department in the Courant Institute of Mathematical Sciences at New York University, New York, NY 10012 USA. His research interests include media forensics, biometrics, misinformation. Memon earned his Ph.D. in computer science from the University of Nebraska. He is an IEEE Fellow, an IAPR Fellow, and an SPIE Fellow. Contact him at memon@nyu.edu.

MOHAN S. KANKANHALLI is a professor at the National University of Singapore, Singapore 119077, Singapore. His research interests include multimedia computing, computer vision, and trustworthy AI. Kankanhalli earned his Ph.D. in computer science from the Rensselaer Polytechnic Institute. He is an IEEE Fellow, an ACM Fellow, an IAPR Fellow, and a Fellow of the Singapore National Academy of Science. Contact him at dcsmk@nus.edu.sg.

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